F. Ordering Binary Tree

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

you are given an array ***A*** of size ***N*** in **increasing** order. Find an order of these N integers such that, if these integers are inserted into a Binary Search Tree (BST) one by one, the height of the resulting BST is minimized.

A Binary Search Tree is a binary tree in which each node has at most two children, referred to as the left and right child. For any node, all elements in the left subtree are smaller than the node's value, and all elements in the right subtree are greater than the node's value.

The height of a Binary Search Tree is defined as the maximum depth among all the nodes in the tree.

Note: All the elements in the array A are guaranteed to be unique. In other words, $A_i \neq A_j$ if $i \neq j$.

Input

The first line contains an integer ***N*** $(1 \leq N \leq 10^5)$ — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq 10^9)$ in non-descending order separated by spaces.

Output

Output the order of the elements such that when inserted into a Binary Search Tree, the height of the tree is minimized. If there are multiple such orders then find any of them.

Example

input Copy

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5
1 2 3 4 5
```

output Copy

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3 1 2 4 5
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